College of Computing

Lab: Composition & LVTI & classes
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Major : Computer Science (CS1)

Learning outcomes:

- Understand composition and inheritance
- Understand local variable type inference
- Translate basic class diagrams into java classes
- Understand Java I/O

Problem 1:

You must create five classes with associated member variables and methods.

The five classes should be created as follows:

1) Create a class and name it Lamp. Inside this class should be declared three member variables: style of type String, battery of type boolean, and globRating of type int. All variables should be marked private. A constructor needs to be created which accepts the three member variables as parameters. In addition, four methods should also be created: turnOn() has no return type and should print a message that the lamp is being turned on; getStyle() returns the lamp style; isBattery returns the value of battery; and getGlobRating() returns the globRating of the lamp.

2) Create a class and name it Bed. Five private member variables should be declared: style of type String, and pillows, height, sheets, quilt of type int. A constructor should be coded which accepts these five member variables as parameters. Also, six methods should be defined: make() has no return type and prints a message to the effect that the bed is being made; getStyle() which returns the value of style; getPillows() returns the number of pillows; getHeight() returns the height of the bed; getSheets() returns the number of sheets on the bed; and getQuilt() returns the value of quilt.

3) Create a class with the name Ceiling. There are two member variables for this one, height and paintedColor, both of type int. There should also be a constructor which accepts both member variables as parameters. There are also two additional methods which should be defined: getHeight() shall return the value of height and getPaintedColor() should return the value of paintedColor.

4) Create a class with the name Wall. It contains one member variable, direction, and is of type String. A constructor for Wall should accept one parameter for the member variable direction. A getter should also be defined for the direction field called getDirection().

5) Create a class with the name Bedroom. This class contains eight member variables:
name of type String; wall1, wall2, wall3, wall4 of type Wall; ceiling of type Ceiling; bed of type Bed, and lamp of type Lamp. The class constructor should accept all eight of the member variables as parameters, and there should also be two additional methods:
getLamp() which returns an object of type Lamp, and makeBed() which prints a message that the bed is being made and also calls the make() method in the Bed class.

Input/Output:

Once you have completed coding your classes you should then use the following code in your main class to test your code and for correct output.

```
1.  Wall wall1 = new Wall("West");
2.  Wall wall2 = new Wall("East");
3.  Wall wall3 = new Wall("South");
4.  Wall wall4 = new Wall("North");
5.
6.  Ceiling ceiling = new Ceiling(12, 55);
7.
8.  Bed bed = new Bed("Modern", 4, 3, 2, 1);
9.
10. Lamp lamp = new Lamp("Classic", false, 75);
11.
12. Bedroom bedRoom = new Bedroom("YOUR NAME HERE", wall1, wall2, wall3,
    wall4, ceiling,bed, lamp);
13. bedRoom.makeBed();
14.
15. bedRoom.getLamp().turnOn();
```

Problem 2:

Write a program that grades arithmetic quizzes as follows:

1. Ask the user how many questions are in the quiz.
2. Ask the user to enter the key (that is, the correct answers). There should be one answer for each question in the quiz, and each answer should be an integer. They can be entered on a single line, e.g., 34 7 13 100 81 3 9 10 321 12 might be the key for a 10-question quiz. You will need to store the key in an array.
3. Ask the user to enter the answers for the quiz to be graded. As for the key, these can be entered on a single line. Again there needs to be one for each question. Note that these answers do not need to be stored; each answer can simply be compared to the key as it is entered.
4. When the user has entered all of the answers to be graded, print the number correct and the percent correct.

When this works, add a loop so that the user can grade any number of quizzes with a single key. After the results have been printed for each quiz, ask "Grade another quiz? (y/n)."

Problem 3:

Write a program that prompts the user for an integer, then asks the user to enter that many values. Store these values in an array and print the array. Then reverse the array elements so that the first element becomes the last element, the second element becomes the second to last element, and so on, with the old last element now first. Do not just reverse the order in which they are printed; actually change the way they are stored in the array. Do not create a second array; just rearrange the elements within the array you have. (Hint: Swap elements that need to change places.) When the elements have been reversed, print the array again.

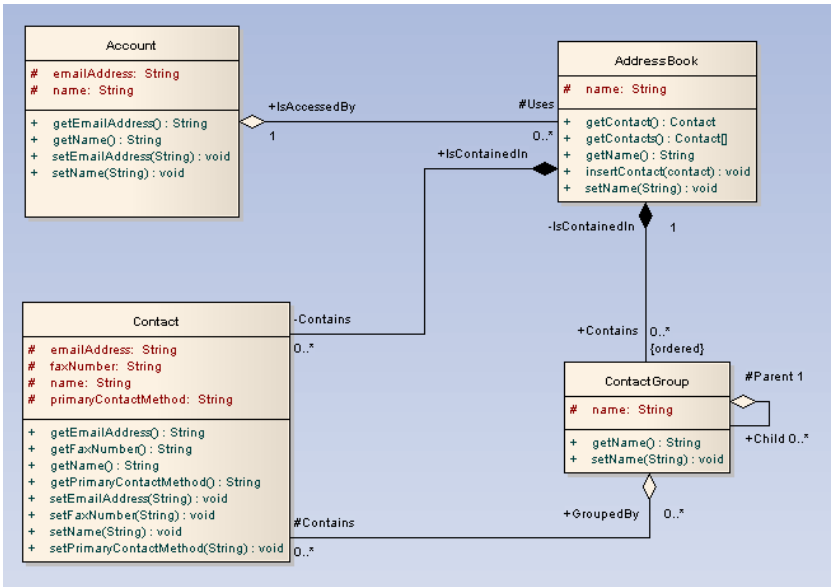
Problem 4:

You will complete a class that implements a shopping cart as an array of items. The file Item.java contains the definition of a class named Item that models an item one would purchase. An item has a name, price, and quantity (the quantity purchased). The file ShoppingCart.java implements the shopping cart as an array of Item objects.

1. Complete the ShoppingCart class by doing the following:
   a. Declare an instance variable cart to be an array of Items and instantiate cart in the constructor to be an array holding capacity Items.
   b. Fill in the code for the increaseSize method. Try to increase size by 50%
   c. Fill in the code for the addToCart method. This method should add the item to the cart and update the totalPrice instance variable (note this variable takes into account the quantity).
   d. Compile your class.

2. Write a program that simulates shopping. The program should have a loop that continues as long as the user wants to shop. Each time through the loop read in the name, price, and quantity of the item the user wants to add to the cart. After adding an item to the cart, the cart contents should be printed. After the loop print a "Please pay ..." message with the total price of the items in the cart.

Problem 5:



Implement the following class diagram using java.

Add the following methods:

- Account
  - public AddressBook[] getAddressBooks();

    - public void addAddressBook(AddressBook book);
    - public void printAllContactDomains();
    - public String getFullName() – "firstName lastName", trimmed.
- AddressBook
    - public Contact[] getContacts();
    - public void addContact(Contact c);
    - public String getSummary();
    - public Contact findByEmail(String email);
- Contact
    - public String getEmailDomain();
    - public String getContactSummary();
- ContactGroup
    - public void addContact(Contact c);
    - public String buildGroupReport(); (try to use StringBuilder class)